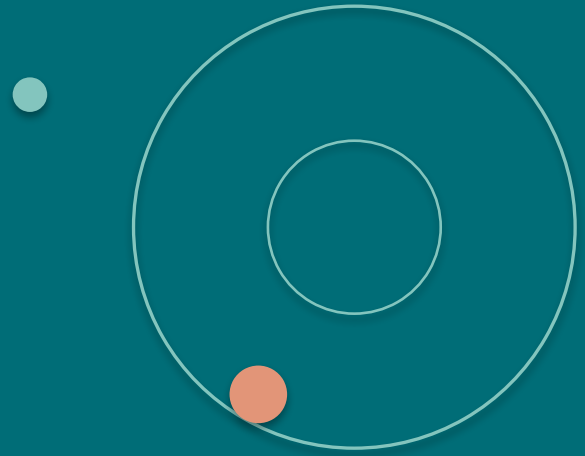
Formalization • Lean 4 + Mathlib

The Identifiability Limit

How much of this work is human?

Now 153 machine-checked theorems.



153
proved theorems

0
sorry, added axioms

12
formalized modules

CONTENTS

What This Deck Covers

01 **The Bird You Miss** — Not detected is not the same as absent

02 **The Detection Ceiling** — Effort never passes that ceiling

03 **Truth Is Not a Point** — The protocol decides what you can see

04 **The Sign Is Unidentified** — One estimator cannot fix its own sign

05 **Claims Become Theorems** — 153 Lean 4 proofs, zero sorry, zero added axioms

01

The Bird You Miss

Ecologists said this long before AI.

ECOLOGY • AN OLD STORY

The Bird You Never See

Unseen ≠ Absent

- The bird is not in this stand
- It is there, but too dark, too dense
- It is there, but wind and distance

Repeated surveys raise detection, never to 1.
The ceiling is set by the protocol.



Under one protocol, look-alike states are lumped together. More sampling raises detection; what the protocol cannot separate, it never will.

THE PROBLEM • ONE MISTAKE

AI Repeats This Mistake

Ecologists	AI-assisted research today
Unseen → recorded as undetected	Undetected → taken as absent
Reference: the true species distribution	Reference: the published record
Change protocol → conclusion changes	Change model → conclusion flips
Repeat surveys → detection rises	More rounds → actually more accurate?

AI research treats model output as the observation itself.

02

The Detection Ceiling

Not new. A theorem from decades ago, rarely pushed.



CONTENT LAYER • RESULT 1

The Best Detector Is Already Computed

TV / 2

Best any detector beats a coin flip

(1 – TV) / 2

Smallest error rate of the best detector

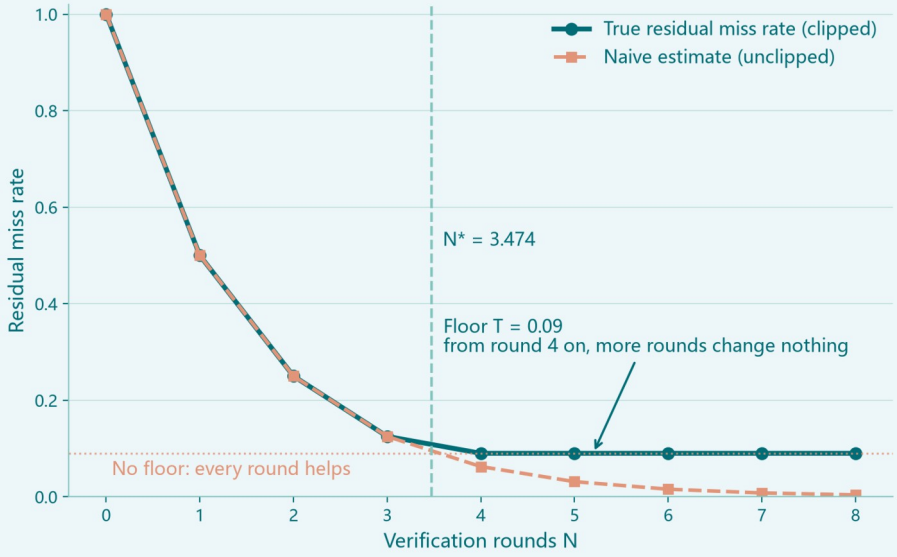
TV = how far apart two distributions are, 0 to 1. TV = 0 means you cannot tell them apart.

Example: reference p=(0.42, 0.28, 0.18, 0.12), model q=(0.10, 0.20, 0.40, 0.30)
TV = 0.40 → at best 0.20 above chance, error rate 0.30.

Source: Fig. 1 — the Neyman-Pearson / Le Cam two-point lemma, not our result

CONTENT LAYER • RESULT 2

More Verification Is Not Better



Parameters: TV = 0.09, q = 0.5, λ = 1 (as in Fig. 2)

Miss rate falls, then stops

- Each round multiplies the miss probability: geometric decay
- But there is a floor it never falls below
- Floor = λ × TV: how separable the two are

Floor T = 0.09: more reviewers stop here

CONTENT LAYER • RESULT 2 (CONT.)

Where the Budget Sticks: Three Numbers

N_c

3·793

Set by reviewer reliability

N_0

3·474

Hard cap, no human in it

N*

3·474

The smaller: here N_0

Perfect reviewers still cap the budget at 3.474.
The remaining misses no headcount can recover.

- **In 83.97% of parameter sets the detector binds, not the person.**

Parameters: TV = 0.09, q = 0.5, λ = 1, as in Fig. 2 (N_c: another reviewer p)

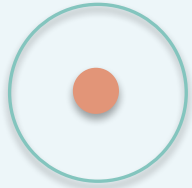
03

Truth Is Not a Point

Change the protocol and the truth moves.

SYSTEM LAYER • CONCEPT

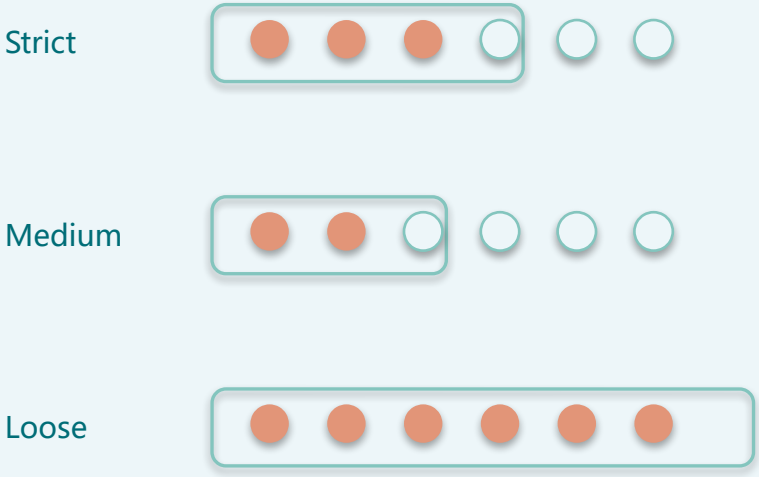
Truth Is a Class, Not a Point



Old way: the answer is one point; estimate the distance.

Our way: which states look identical under this protocol?

They form a class [s*]. Error becomes a function of the protocol.



The same truth mixes different states under different protocols.

SYSTEM LAYER • RESULT 3

Two Accounts of Credit

Human credit: change the layer

- Ask a better question
- Formally: change Φ(α) itself
- Gain has no ceiling

AI credit: compress within the layer

- Push error toward Φ(α)
- Marginal gain shrinks, hard cap



AI hits a wall inside the layer; a human changes layer.

"AI helps" holds only when the layer is fixed, the state with no human credit.

SYSTEM LAYER • REBUTTAL

Influence ≠ Value

$+M$

$-M$

Two outputs, identical |influence|.

One is +M, the other −M, and M is arbitrary.

Influence cannot stand in for value.
That is a defect of definition.



KL and mutual information measure magnitude only.

Value needs a sign.

SYSTEM LAYER • VERDICT

Incommensurable Has Two Regimes

When the floor binds

Incommensurable

Reviewers of any reliability: same budget.
Not even a ranking survives.

When the floor is slack

Commensurable

Who is stronger is visible.
But the rate is not proportional.

The exact claim: not linearly convertible.
The stronger version is refuted; formalization rewrote our claim.

04

Sign Is Not Identified

The problem is direction, not precision.

CALIBRATION LAYER • PROBLEM

The Standard Estimator Has No Human

$$\hat{\varphi} = 1 - \bar{\tau} / \bar{s}$$

$\bar{\tau}$ = the fitted plausibility bar

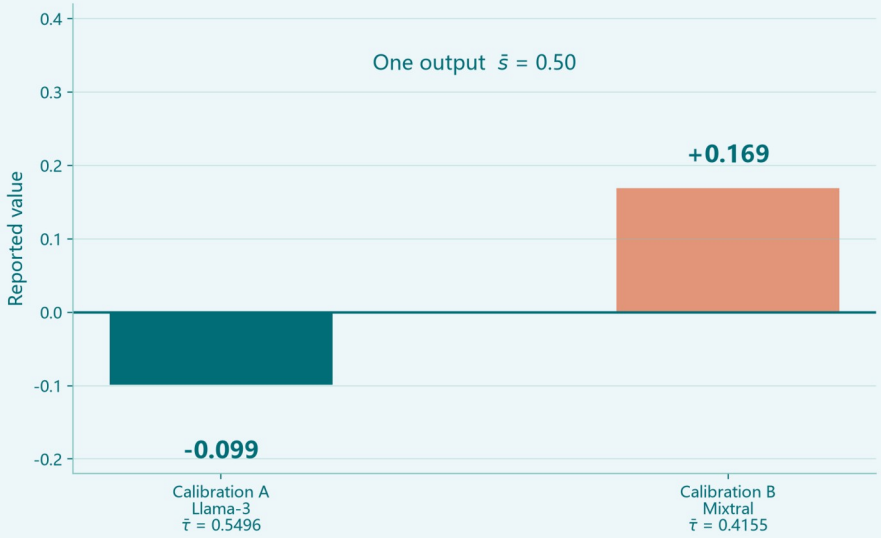
$\bar{s}$ = mean surprisal of the output

The human input x is not in the formula.

It measures how surprising the output is, not how human.

CALIBRATION LAYER • THEOREM

One Output, Two Signs



Fix the output: $\bar{s}$ = 0.50

Calibration A • Llama-3: $\bar{\tau}$ = 0.5496
Reports −0.099: almost no human

Calibration B • Mixtral: $\bar{\tau}$ = 0.4155
Reports +0.169: human present

Same text, same data, opposite verdicts.

Only the bar moved, and it was fitted on 2500 unrelated chat prompts.

$\bar{\tau}$ from Xie et al. (ACL 2026), Appendix A.5

CALIBRATION LAYER • CONSEQUENCE

It Is Not Even a Proportion

Bounded above

$\hat{\varphi} < 1$ always.

More surprising → closer to 1.

No lower bound

$\hat{\varphi}$ goes below 0.

More predictable → further down.

Reading it as a proportion in [0,1] fails in the estimator.

The authors saw these negatives and explained them inside their own framework:

"could plausibly have been generated with minimal or negligible human contribution" — Xie et al., §5.5

05

Claims Become Theorems

153 theorems, zero sorry, zero added axioms.

FORMALIZATION

153 Theorems, 12 Modules

Total variation
Basic

Contamination
Contamination

Coverage
Coverage

Hypothesis testing
HypothesisTest

Refinement and observation
Refinement

Influence and value
Value

Verification budget
VerifyBudget

Incommensurability
Incommensurable

Unidentified calibration
Calibration

Fragmentation
Fragmentation

Corpus as selection
Corpus

Package root
Provenance

Lean 4 v4.15.0 + Mathlib • 153 theorems + 20 worked examples • zero sorry, zero added axioms

WHAT THE FORMALIZATION WAS WORTH

It Caught Us First

An inequality was backwards

The proof failed. Lean killed our own modelling error.

The residual had no floor

"More verification is better" refuted; miss rate has a floor.

Two monotonicities reversed

We expected the budget to fall with TV. It does not.

"Fully incommensurable" too strong

Forced to the exact claim: not linearly convertible.

A modelling error that survives the prover will not survive a coauthor.

CLOSING

What We Proved

- Unseen ≠ absent. AI treats model output as the observation.
- Detection has a ceiling. Effort does not move it.
- The verification budget has a cap with no human in it.
- Influence does not measure value. Value needs a sign.
- The sign is not identified either; an out-of-distribution bar decides it.
- 153 theorems, zero sorry, zero added axioms, reproducible build.

We do not attack existing measures; they work for their stated purpose.
We turned their prose warnings into theorems a machine can check.